Companion Appendix to "When Do Local Income Taxes Cause Residential Sorting? Evidence from Pennsylvania's Piecewise EIT System"

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This appendix collects supplementary results for the main paper. The main paper's preferred causal estimate is the block-level binding-state decomposition: a 1 percentage-point municipal resident-rate increase in the Inc-bind cell reduces the log count of workers living in the block by roughly -0.21 log points, with a clean realized-dose pre-trend and a null Inc-nobind within-design placebo. Municipality-level results are used for policy-jurisdiction magnitude calibration and for the industry mechanism; the municipality-level Inc-bind coefficient is larger but the realized-dose pre-trend fails parallel trends, so the municipality-level estimate is not the causal headline. School-district-level results are reported as aggregation diagnostics and are not load-bearing for the main causal claim; they do not robustly detect a sorting response. Block-level share outcomes are reported as supporting estimates and are fragile at block geography.

Section, table, and figure numbering in this appendix is self-contained.

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1 Municipality-Level Full Inc/Dec Detail

Full Inc-cell coefficients with diagnostics

The remainder of this subsection retains the prior detailed municipality-level headline analysis (Inc-cell coefficients with full diagnostics from Equation **??**) for completeness.

Table 1 reports the muni-resident-rate Inc-cell coefficients across the eight outcomes. Each row is a separate regression of the form in Equation (**??**); the displayed columns are the muni-side Inc-bind, Inc-becomes, and Inc-nobind cells. SD and nonresident-rate cell coefficients are estimated jointly in the same regression and are reported in Table **??** (channel separation). Dec-cell coefficients are in Table 2.

Table 1: Headline stacked-continuous DiD: muni-rate Inc cells with diagnostics

Outcome	Inc-bind (n_{muni}=40)	Inc-becomes (n_{muni}=58)	Inc-nobind (placebo) (n_{muni}=26)
Coefficients (per 1 pp of $\Delta\tau^{muni}$)			
log(live_and_work)	-0.402^{***}	-0.181^{**}	-0.062
	(0.096)	(0.072)	(0.169)
log(in_commuters)	-0.213^{**}	-0.197^{***}	-0.187
	(0.083)	(0.054)	(0.204)
log(out_commuters)	-0.053	-0.042	$+0.217^{**}$
	(0.038)	(0.029)	(0.091)
log(total_jobs)	-0.270^{***}	-0.212^{***}	-0.201
	(0.086)	(0.054)	(0.203)
log(total_residents)	-0.100^{***}	-0.068^{*}	$+0.194^{**}$
	(0.033)	(0.040)	(0.095)
share_live_in	-0.045^{***}	-0.012	-0.012
	(0.010)	(0.009)	(0.008)
share_work_outside	$+0.041^{***}$	$+0.020^{***}$	$+0.017$
	(0.016)	(0.007)	(0.014)
Diagnostics (log_live_and_work regression)			
N	218,685		
R^2	0.964		
Within R^2	0.0007		
Dep. var. mean (log_live_and_work)	3.677		
Dep. var. SD	1.934		
Fixed effects	stack-muni + stack-year		
Clusters (muni level)	1,945		

Notes: Stacked continuous DiD, Equation (**??**), drop-incomplete sample with NC controls strictly defined as munis with no rate change on the relevant side. Each muni in a treated cell appears in one sub-experiment (its own first-event cohort); NC munis appear in every sub-experiment. Only muni-side Inc cells are displayed here; SD and nonresident-side cells are estimated jointly in the same regression and reported in Tables **??** and 4. Each regression also includes the share-based control vector (police, streets, debt service / total expenditures; EIT, real estate, intergov / total revenues);

Inc-bind: the binding-cell effect. A 1 percentage-point increase in the muni resident rate in the Inc-bind cell (where the bill rises by the full $\Delta\tau$ for in-muni residents) reduces `live_and_work` by roughly 33 percent (a -0.40 log-point coefficient), `in_commuters` by roughly 19 percent (-0.21 log points), `total_jobs` by roughly 24 percent (-0.27 log points), and `total_residents` by roughly 10 percent (-0.10 log points). The dependent variable mean on `log(live_and_work)` is 3.68, so the typical muni has $e^{3.68} \approx 40$ live-and-work workers; a 1 pp hike at this mean predicts $40 \cdot (e^{-0.402} - 1) \approx -13$ live-and-work workers. Actual rate hikes in the Inc-bind cell average 0.27 pp, which scales the predicted effect down to roughly $0.27 \cdot 40 \cdot (e^{-0.402} - 1) \approx -3.6$ workers per typical event over 5 years.

Inc-becomes: the bill-flip cell. Inc-becomes events show smaller but still significant effects: -0.181 on `live_and_work`, -0.212 on `total_jobs`, -0.068 on `total_residents`. The attenuation relative to Inc-bind is consistent with the bill rising by less than the full $\Delta\tau$ in this cell—only the portion of the rate change above the work nonresident rate is incident on the worker.

Dec cells: thin and noisy. Dec-cell coefficients (Table 2) are reported for completeness but are not robust to specification. Dec-bind has 11 munis; Dec-becomes-non has 4; Dec-nobind has 15. We do not draw asymmetric-response conclusions from these cells.

Table 2: Headline stacked-continuous DiD: muni-rate Dec cells (thin)

Outcome	Dec-bind (n_{muni}=11)	Dec-becomes-non (n_{muni}=4)	Dec-nobind (n_{muni}=15)
log(live_and_work)	$+0.241^{***}$ (0.078)	-0.260 (0.272)	$+0.332^{**}$ (0.165)
log(in_commuters)	$+0.052$ (0.080)	$+0.496$ (0.339)	$+0.013$ (0.121)
log(out_commuters)	-0.068 (0.061)	-0.628 (0.416)	$+0.132$ (0.101)
log(total_jobs)	$+0.111$ (0.073)	$+0.420$ (0.265)	$+0.022$ (0.116)
log(total_residents)	-0.056 (0.043)	-0.651 (0.420)	$+0.135$ (0.099)
share_live_in	$+0.054^{*}$ (0.030)	-0.071 (0.095)	$+0.012$ (0.030)
share_work_outside	-0.011 (0.013)	$+0.014$ (0.017)	-0.003^{*} (0.001)

Notes: Same regression as Table 1. Dec cells are thin and noisy; we do not draw causal conclusions from these coefficients individually. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Per-outcome diagnostics. Table 3 reports N, R^2, within-R^2, and the dependent-variable mean for each headline regression.

Table 3: Headline regressions: per-outcome diagnostics

Outcome	N	R^2	Within R^2	Dep. mean	Dep. SD
log(live_and_work)	218,685	0.964	0.0007	3.677	1.934
log(in_commuters)	239,031	0.959	0.0011	5.847	1.887
log(out_commuters)	242,730	0.979	0.0009	6.636	1.263
log(total_jobs)	239,041	0.961	0.0011	5.948	1.901
log(total_residents)	242,730	0.981	0.0010	6.697	1.299
share_live_in	239,041	0.823	0.0010	0.093	0.076
share_work_outside	242,730	0.945	0.0008	0.943	0.066
share_nonres	239,041	0.823	0.0010	0.907	0.076

Notes: Same regression structure as Table 1. R^2 is the standard within-stack coefficient of determination; within-R^2 is the variation explained by the cell×dose interactions and controls, net of the muni-stack and year-stack fixed effects. Dep. mean and SD computed over the regression sample.

Full main specification with control estimates. Table 4 reports all coefficients from the headline regression on `log(live_and_work)`: muni-side cells, SD-side cells, nonresident-side cells, and control coefficients.

Table 4: Full main spec: log(live_and_work) on all treatments + controls

Term	Coefficient	SE
Muni resident-rate cells (vs NC^{muni})		
Inc-bind $\times\Delta\tau^{\text{muni}}$	-0.402^{***}	0.096
Inc-becomes $\times\Delta\tau^{\text{muni}}$	-0.181^{**}	0.072
Inc-nobind $\times\Delta\tau^{\text{muni}}$	-0.062	0.169
Dec-bind $\times\Delta\tau^{\text{muni}}$	$+0.241^{***}$	0.078
Dec-becomes-non $\times\Delta\tau^{\text{muni}}$	-0.260	0.272
Dec-nobind $\times\Delta\tau^{\text{muni}}$	$+0.332^{**}$	0.165
School district rate cells (vs NC^{SD})		
Inc-bind $\times\Delta\tau^{\text{SD}}$	$+0.177^{*}$	0.099
Inc-becomes $\times\Delta\tau^{\text{SD}}$	$+0.103$	0.090
Dec-bind $\times\Delta\tau^{\text{SD}}$	-0.721^{***}	0.191
Dec-becomes-non $\times\Delta\tau^{\text{SD}}$	-0.223^{**}	0.109
Dec-nobind $\times\Delta\tau^{\text{SD}}$	$+0.048$	0.179
Share-based controls		
share_police	$+0.097$	0.100
share_streets	$+0.049$	0.039
share_debt_svc	-0.058	0.051
share_eit	-0.022	0.085
share_re	-0.182^{**}	0.087
share_intergov	$+0.016$	0.044
Diagnostics		
N	218,685	
R^2	0.964	
Within R^2	0.0007	
Dep. mean (log_live_and_work)	3.677	
Dep. SD	1.934	
Fixed effects	stack-muni + stack-year	
Clusters (muni)	1,945	

Notes: All coefficients from a single regression of log(live_and_work) on cell-specific dose interactions

2 Municipality-Level Heterogeneity by Muni Type

Municipality-Level Heterogeneity by Muni Type

The headline municipality-level coefficient is concentrated in 8 post-industrial PA cities (Wilkes-Barre, Hazleton, Johnstown, Jeannette, Coatesville, Warren, Carbondale, Farrell). Table 5 reports the muni-Inc-bind cell coefficient on `log(live_and_work)` by muni type.

Table 5: Inc-bind cell by muni type (drop sample, post-NC-fix)

Type	n	Mean residents	Inc-bind coef	Sig
City	8	6,904	−0.455	** (p=0.021)
Non-city (borough + township)	32	3,207	−0.194	— (p=0.189)
Borough	10	2,631	−0.213	— (p=0.173)
Township	22	3,468	−0.161	— (p=0.438)
Pooled (headline)	40	—	−0.402***	—

Notes: Stacked-continuous DiD with same controls and FE structure as the municipality-level headline. Each row restricts the sample to its own muni type; NC controls are restricted to the same type. Cluster-robust SEs by muni. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

The 8 cities drive the municipality-level headline. The 32 borough and township Inc-bind munis show direction-consistent but individually insignificant effects.

3 Block-Level Share Outcomes (Supporting)

The main paper reports block-level count outcomes (`log_home_count`, `log_work_count`) as the primary residential-sorting evidence. Block-level share outcomes are reported here for completeness. Block geography has small and sometimes sparse denominators, and both the numerator and denominator can move with the same residential decision, which makes share estimates mechanically fragile at this scale. I do not use the share-outcome coefficients to

identify the headline causal effect; the count coefficients in the main text are the preferred estimands.

Table 6: Block-level dose-response: muni resident-rate cells (share outcomes, supporting)

Outcome	Share live+work in muni	Share work outside
Inc-bind	-0.035	0.035
	(0.031)	(0.031)
Inc-becomes	-0.043***	0.043***
	(0.010)	(0.010)
Inc-nobind	-0.037**	0.037**
	(0.015)	(0.015)
Dec-bind	-0.049	0.049
	(0.031)	(0.031)
Dec-becomes-non	-0.080**	0.080**
	(0.033)	(0.033)
Dec-nobind	0.006	-0.006
	(0.006)	(0.006)
N	23,567,655	23,567,655
R^2	0.563	0.563
Within R^2	0.000	0.000
Dep. mean	0.121	0.879
Dep. SD	0.184	0.184

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with block-by-stack and year-by-stack fixed effects; standard errors two-way clustered at the muni and SD levels. Each block is exposed to one municipality and one school district; both rate-setters' policy variation enters at its native level. Block-level share outcomes are reported for completeness. Sparse denominators and endogenous denominator dynamics make these estimates mechanically fragile at block geography; they are not used to identify the headline causal effect. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 7: Block-level dose-response: SD resident-rate cells (share outcomes, supporting)

Outcome	Share live+work in muni	Share work outside
Inc-bind	0.006*	-0.006*
	(0.003)	(0.003)
Inc-becomes	-0.002	0.002
	(0.006)	(0.006)
Dec-bind	-0.018**	0.018**
	(0.007)	(0.007)
Dec-becomes-non	-0.007***	0.007***
	(0.003)	(0.003)
Dec-nobind	-0.034**	0.034**
	(0.017)	(0.017)
N	23,567,655	23,567,655
R^2	0.563	0.563
Within R^2	0.000	0.000
Dep. mean	0.121	0.879
Dep. SD	0.184	0.184

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with block-by-stack and year-by-stack fixed effects; standard errors two-way clustered at the muni and SD levels. Each block is exposed to one municipality and one school district; both rate-setters' policy variation enters at its native level. Block-level share outcomes are reported for completeness; not used for headline identification. Rows omitted (no estimable coefficients in any outcome column): Inc-nobind. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

4 School-District-Level Results

School-District-Level Headline Results

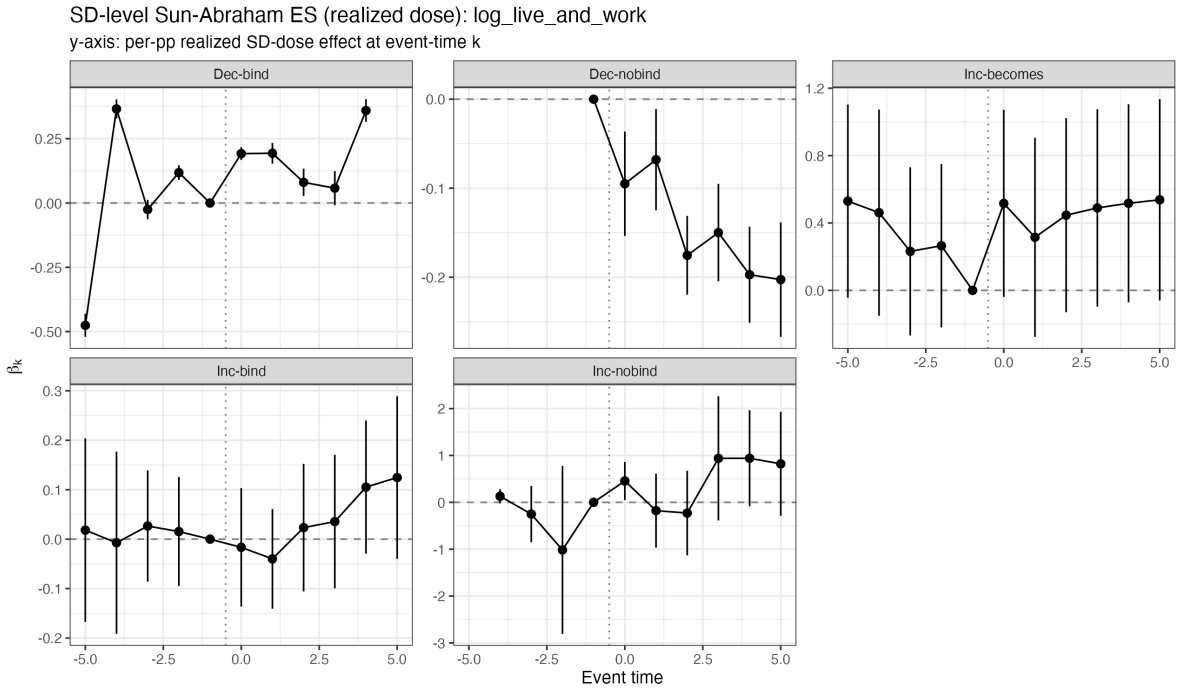
Table **??** reports the SD-level dose-response coefficients. The unit of observation is SD-year. The SD's own resident rate enters the regression with binding-cell decomposition; the within-SD aggregate of constituent muni rates enters as a control; and SD-level fiscal share controls constructed from Census F-33 data control for SD fiscal posture. Binding-state classification at the SD level uses the within-SD mean of `muni_nonres_eit` (the work-side nonresident rate) as the comparator, aggregating the institutional rate field across the SD's constituent munis (the previous draft used the within-SD mean resident rate as a proxy; that proxy has been retired and the cell labels reproduce essentially unchanged under the corrected definition).

Table 8: School-district-level dose-response by binding-state cell

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind	0.027	0.057	0.106***	0.003	0.000
	(0.038)	(0.046)	(0.030)	(0.004)	(0.004)
Inc-becomes	0.271*	0.184	0.318**	0.004	0.012
	(0.154)	(0.137)	(0.131)	(0.006)	(0.010)
Inc-nobind	0.793	1.149*	0.422***	-0.035	-0.015
	(0.506)	(0.633)	(0.148)	(0.023)	(0.021)
Dec-bind	0.060	0.050	0.178***	0.000	0.006***
	(0.077)	(0.111)	(0.045)	(0.004)	(0.002)
Dec-nobind	-0.126***	0.154***	0.076***	-0.032***	0.021***
	(0.025)	(0.028)	(0.016)	(0.002)	(0.003)
N	51,420	51,454	51,454	51,454	51,454
R^2	0.986	0.987	0.987	0.953	0.977
Within R^2	0.004	0.004	0.005	0.004	0.004
Dep. mean	6.260	8.335	8.769	0.125	0.900
Dep. SD	1.298	1.202	0.909	0.068	0.076

Notes: Cluster-robust standard errors in parentheses. Continuous stacked-DiD with SD-by-stack and year-by-stack fixed effects; standard errors clustered at the school-district level. Unit of observation is SD×year. Binding-cell classification uses the within-SD mean of $\tau^{\text{nr,muni}}$ as the work-side comparator, aggregating the institutional nonresident-rate field over the SD's constituent munis. Rows omitted (no estimable coefficients in any outcome column): Dec-becomes-non. These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Figure 1: SD-level realized-dose ES on `log(live_and_work)` by cell.



Notes: per-pp realized-dose coefficient at each event-time, SD-level stacked-continuous DiD. The SD-level results are reported for completeness. They show that sorting is not robustly detected at the school-district aggregation: the SD-resident-rate Inc-bind coefficient is positive in this aggregation, sign-reversed relative to the muni- and block-level findings, and the pre-event lead is marginally non-zero on `log(live_and_work)` ($\beta_{\text{pre}} = +0.061$, $|t| = 1.74$). These estimates should not be interpreted as standalone evidence of an SD-channel residential response.

5 Municipality- and SD-Level Pre/Post Act 32 Splits

6 Realized-Dose Event Study Coefficient Detail

This appendix collects the per-cell post/pre ATT summaries of the realized-dose Sun–Abraham event study described in Section **??**. The visual versions appear in the main paper: Figure **??** (pooled, block); Figure **??** (pooled, muni vs SD); Figures **??**–**??** (block decomposition by cell). The tables below report the numerical post-event mean ATT and pre-event lead by binding-state cell.

Table 9: Realized-dose ES — municipality level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind (post)	-0.414***	-0.229***	-0.100***	-0.045***	0.047***
	(0.071)	(0.066)	(0.029)	(0.009)	(0.010)
Inc-bind (pre)	0.149***	0.043	0.056	0.016	-0.020***
	(0.056)	(0.061)	(0.059)	(0.012)	(0.005)
Inc-becomes (post)	-0.137***	-0.198***	-0.032**	-0.009**	0.009***
	(0.029)	(0.033)	(0.016)	(0.004)	(0.002)
Inc-becomes (pre)	0.055	-0.006	0.068**	0.004	-0.010**
	(0.043)	(0.083)	(0.029)	(0.008)	(0.004)
Inc-nobind (post)	-0.138***	-0.379***	0.057	0.006	0.017***
	(0.040)	(0.109)	(0.045)	(0.006)	(0.006)
Inc-nobind (pre)	0.169**	0.062	-0.042	0.032**	-0.016***
	(0.070)	(0.127)	(0.148)	(0.013)	(0.006)
Dec-bind (post)	0.650***	0.245**	0.185***	0.012	0.004
	(0.185)	(0.098)	(0.041)	(0.012)	(0.005)
Dec-bind (pre)	0.206	-0.073	0.180***	0.006	0.004
	(0.221)	(0.104)	(0.037)	(0.007)	(0.007)
Dec-becomes-non (post)	1.497***	-0.236***	0.500***	0.127***	-0.024***
	(0.065)	(0.065)	(0.030)	(0.004)	(0.002)
Dec-becomes-non (pre)	0.102	-1.425***	-0.311***	0.086***	-0.022***
	(0.084)	(0.088)	(0.042)	(0.007)	(0.003)
Dec-nobind (post)	-0.328**	-0.006	-0.079	-0.059***	0.021***
	(0.130)	(0.069)	(0.070)	(0.015)	(0.008)
N	198,025	216,226	219,444	216,236	219,444

Notes: Cluster-robust standard errors in parentheses. Each cell reports two rows: post-event ATT (mean of β_k for $k = 0..5$) and pre-event lead (mean of β_k for $k \leq -2$). Rows omitted (no estimable coefficients in any outcome column): Dec-nobind (pre). These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 10: Realized-dose ES — school district level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind (post)	0.039	0.066***	0.099***	-0.001	0.000
	(0.027)	(0.022)	(0.019)	(0.002)	(0.002)
Inc-bind (pre)	0.013	0.015	-0.015	-0.008	0.001
	(0.039)	(0.031)	(0.025)	(0.005)	(0.003)
Inc-becomes (post)	0.470***	0.311***	0.426***	0.010***	-0.002
	(0.121)	(0.105)	(0.093)	(0.003)	(0.004)
Inc-becomes (pre)	0.372***	0.225*	0.193*	0.013	-0.027***
	(0.139)	(0.119)	(0.105)	(0.009)	(0.009)
Inc-nobind (post)	0.458**	0.934***	0.302***	-0.059***	-0.015
	(0.202)	(0.239)	(0.060)	(0.015)	(0.011)
Inc-nobind (pre)	-0.379	-0.224	-0.171	-0.036**	-0.008
	(0.322)	(0.290)	(0.144)	(0.016)	(0.007)
Dec-bind (post)	0.177***	-0.245***	0.411***	0.062***	0.013***
	(0.011)	(0.011)	(0.006)	(0.001)	(0.001)
Dec-bind (pre)	-0.004	-0.601***	0.200***	0.086***	0.010***
	(0.010)	(0.009)	(0.006)	(0.001)	(0.001)
Dec-nobind (post)	-0.148***	0.147***	0.068***	-0.034***	0.024***
	(0.012)	(0.012)	(0.008)	(0.002)	(0.001)
N	51,128	51,162	51,162	51,162	51,162

Notes: Cluster-robust standard errors in parentheses. Each cell reports two rows: post-event ATT (mean of β_k for $k = 0..5$) and pre-event lead (mean of β_k for $k \leq -2$). Rows omitted (no estimable coefficients in any outcome column): Dec-becomes-non (post), Dec-becomes-non (pre), Dec-nobind (pre). These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 11: Realized-dose ES — block level (muni cohort)

Outcome	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Inc-bind (post)	-0.305***	-0.079*	-0.043***	0.043***
	(0.041)	(0.048)	(0.014)	(0.014)
Inc-bind (pre)	-0.027	0.017	0.020**	-0.020**
	(0.053)	(0.033)	(0.008)	(0.008)
Inc-becomes (post)	0.009	-0.046***	-0.030***	0.030***
	(0.020)	(0.015)	(0.006)	(0.006)
Inc-becomes (pre)	0.080***	0.032**	0.003	-0.003
	(0.026)	(0.014)	(0.007)	(0.007)
Inc-nobind (post)	0.045**	-0.050***	-0.042***	—
	(0.018)	(0.019)	(0.008)	
Inc-nobind (pre)	0.123***	0.025	0.008***	—
	(0.019)	(0.016)	(0.003)	
Dec-bind (post)	0.278***	0.294***	-0.052***	—
	(0.055)	(0.110)	(0.010)	
Dec-bind (pre)	0.173***	0.457***	0.028**	—
	(0.057)	(0.099)	(0.011)	
Dec-becomes-non (post)	0.823***	-0.815***	-0.119***	0.119***
	(0.037)	(0.085)	(0.011)	(0.011)
Dec-becomes-non (pre)	0.403***	-1.387***	-0.044***	0.044***
	(0.036)	(0.036)	(0.011)	(0.011)
Dec-nobind (post)	-0.095	0.060	0.015	-0.013
	(36.376)	(0.042)	(11.217)	(11.217)
N	21,404,148	7,988,719	21,404,148	21,404,148

Notes: Cluster-robust standard errors in parentheses. Each cell reports two rows: post-event ATT (mean of β_k for $k = 0..5$) and pre-event lead (mean of β_k for $k \leq -2$). Rows omitted (no estimable coefficients in any outcome column): Dec-nobind (pre). These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

7 Binary Sun–Abraham IW Event Study (Municipality and SD)

The binary Sun–Abraham IW event study described in Section **??** complements the realized-dose ES by testing average parallel trends under a binary cohort indicator rather than a continuous treatment-magnitude tag. Block-level binary SA-IW is unstable under stacking (thin cohort $\times$ event-time cells) and is not reported. The tables below report the post-event mean ATT and pre-event lead by binding-state cell at the municipality and school-district levels.

Table 12: Binary Sun–Abraham IW ES — municipality level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind (post)	-0.014	0.073***	-0.009	-0.013***	0.003
	(0.030)	(0.027)	(0.015)	(0.003)	(0.003)
Inc-bind (pre)	0.110	0.000	0.054	0.006	-0.012**
	(0.076)	(0.044)	(0.041)	(0.007)	(0.005)
Inc-becomes (post)	-0.134***	0.031	-0.010	-0.012***	0.004**
	(0.021)	(0.022)	(0.008)	(0.003)	(0.002)
Inc-becomes (pre)	-0.114***	0.064	0.058***	-0.027***	-0.002
	(0.033)	(0.042)	(0.010)	(0.004)	(0.004)
Inc-nobind (post)	0.180**	0.091	0.021	0.013***	-0.004
	(0.078)	(0.102)	(0.020)	(0.005)	(0.005)
Inc-nobind (pre)	0.113	-0.300**	-0.079***	0.011*	-0.002
	(0.077)	(0.130)	(0.016)	(0.006)	(0.005)
Dec-bind (post)	-0.191***	-0.305***	-0.020	-0.011*	-0.002
	(0.062)	(0.067)	(0.028)	(0.006)	(0.005)
Dec-bind (pre)	-0.253***	-0.038	-0.187***	-0.007*	0.009
	(0.087)	(0.061)	(0.026)	(0.004)	(0.006)
Dec-becomes-non (post)	0.046	0.055	0.010	0.002	0.041
Dec-becomes-non (pre)	0.865	0.140	-0.087	0.058	-0.036
Dec-nobind (post)	-0.388***	-0.393***	-0.057**	0.018**	0.021***
	(0.089)	(0.071)	(0.025)	(0.009)	(0.008)
Dec-nobind (pre)	0.788***	0.174***	0.228***	0.058***	-0.010**
	(0.048)	(0.037)	(0.019)	(0.004)	(0.004)

Notes: Cluster-robust standard errors in parentheses. Binary Sun–Abraham interaction-weighted ES. Post = mean of β_k for $k = 0..5$; pre = mean for $k \leq -2$. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 13: Binary Sun–Abraham IW ES — school district level

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Inc-bind (post)	0.058*	0.030	0.097***	-0.001	0.005*
	(0.031)	(0.024)	(0.023)	(0.002)	(0.003)
Inc-bind (pre)	0.099***	0.110***	0.079***	-0.003	-0.014***
	(0.024)	(0.027)	(0.016)	(0.002)	(0.002)
Inc-becomes (post)	1.226***	0.984***	1.078***	0.038***	-0.017***
	(0.141)	(0.112)	(0.112)	(0.004)	(0.002)
Inc-becomes (pre)	1.111***	1.505***	0.679**	-0.055***	-0.039***
	(0.417)	(0.345)	(0.341)	(0.011)	(0.005)
Dec-bind (post)	0.074	0.035	-0.009	0.004	-0.011
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Dec-bind (pre)	-0.174	0.077	-0.129	-0.028	0.010
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)

Notes: Cluster-robust standard errors in parentheses. Binary Sun–Abraham interaction-weighted ES. Post = mean of β_k for $k = 0..5$; pre = mean for $k \leq -2$. Rows omitted (no estimable coefficients in any outcome column): Inc-nobind (post), Inc-nobind (pre), Dec-becomes-non (post), Dec-becomes-non (pre), Dec-nobind (post), Dec-nobind (pre). These cells contain too few units, or all coefficients were dropped for collinearity with the stack-interacted fixed effects. $^{*}p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

8 Dose-Magnitude Linearity (Municipality, SD, Block Terciles)

9 Alternative Control Group: Clean-in-Window

10 Data Construction Pipeline (Step-by-Step)

This appendix expands the build pipeline summarized in the main paper's Data section. The analysis links two administrative systems that do not share common identifiers: LEHD/LODES employment records (Census block geography) and Pennsylvania EIT rates (DCED municipality IDs). The build proceeds through the following steps.

1. **Scrape PA DCED MunStats tax rates.** PA does not publish a downloadable panel of historical EIT rates. The only source is the DCED MunStats portal, which serves one XLS file (∼12 MB) per year via an ASP.NET form. A Python script submits requests for each year 2000–2026, extracting resident and nonresident EIT rates, school district EIT rates, and Local Services Tax for all ∼2,500 municipalities. About 27% of munis report multiple school-district rates; in the municipality- level panel each municipality-year is assigned the maximum SD rate.

2. **Scrape PA DCED Annual Financial Reports.** Fiscal controls (share-based) are constructed from the DCED AFR (71 columns per municipality-year) for 2002–2023, trimmed at load to 2002–2019: shares of police, streets, debt service in total expenditures and shares of EIT, real estate tax, intergovernmental revenue in total revenues.

3. **Backfill nonresident rates.** The DCED nonresident registry was incomplete pre-Act 32: 55–60% coverage 2002–2011, 72% in 2012, 77% by 2019. Diagnostics show this is a reporting gap, not a regime gap (0.2% of late-reporting munis have a true zero nonresident rate; 96.3% of nonresident rates are perfectly stable across 2012–2019). I backfill missing pre-2012 nonresident rates with the earliest observed post-2012 rate per muni and flag the imputation.

4. **Download LODES block-level OD files.** LODES Version 8 (2020 Census geography retroactively applied to 2002–2021), block-level (`agg_geo = "block"` in the `lehdr` R package). I download main files (intra-state) and auxiliary files (cross-state). All-jobs (S000), earnings (SE01, SE03), and age (SA01–SA03) segments. Reference date: April 1.

5. **Crosswalk PA DCED to Census geography.** DCED municipality names matched to TIGER/Line COUSUBFP via normalized name + county; 99.6% match (9 unmatched munis dissolved or merged before 2020).

6. **Build a block-to-municipality crosswalk.** Point-in-polygon assignment of TIGER 2020 block centroids to PA cousub geometries. Because 2020 Census blocks nest in cousubs by construction, this assignment is exact: every block has an unambiguous municipality. The crosswalk recovers all 2,561 PA municipalities. The alternative (tract-centroid assignment) silently drops 1,082 of 2,561 municipalities (42%) and misattributes employment for the retained ones.

7. **Compute neighbor pairs.** Centroid-to-centroid distances between all PA muni pairs; retain pairs within 80 km.

8. **Aggregate LODES.** Block-pair-year LODES is summed to (work municipality, home municipality) × year, producing total jobs, live-and-work, in-commuters, and cross-state commuters per municipality-year. Block-level outcomes (`home_count`, `work_count`, `home_in_muni_share`) are retained directly on the block-year panel without aggregation.

9. **Merge rates and controls.** Final join: rate panels and fiscal controls onto the OD panels, producing the analysis-ready municipality × year, school- district × year, and block × year datasets.

11 Pooled Inc/Dec Binary-Event Analog

The binary-event analog of the per-pp pooled specification (Equation **??**) replaces each $\Delta\tau_{it}^{s}$ dose interaction with a post-treatment indicator that equals one in the event year and after. The four reported coefficients per regression are $\{\text{everInc}^{M}, \text{everDec}^{M}, \text{everInc}^{SD}, \text{everDec}^{SD}\}$. Sign-wise the binary estimates agree with the per-pp pooled headline at every level.

Table 14: Pooled Inc/Dec headline — block level (binary event)

Outcome	Log(workers live)	Log(workers work)	Share live+work in muni	Share work outside
Municipality rate (τ^M)				
$\Delta\tau > 0$ (Inc)	-0.052**	-0.038**	-0.018**	0.018**
	(0.025)	(0.016)	(0.008)	(0.008)
$\Delta\tau < 0$ (Dec)	-0.050***	-0.028	0.013***	-0.013***
	(0.017)	(0.053)	(0.004)	(0.004)
School district rate (τ^{SD})				
$\Delta\tau > 0$ (Inc)	0.003	-0.000	0.001	-0.001
	(0.006)	(0.007)	(0.001)	(0.001)
$\Delta\tau < 0$ (Dec)	0.008	0.066***	0.005***	-0.005***
	(0.006)	(0.015)	(0.001)	(0.001)
N	23,567,655	8,751,581	23,567,655	23,567,655
R^2	0.858	0.865	0.563	0.563
Within R^2	0.000	0.000	0.000	0.000
Dep. mean	2.238	2.318	0.121	0.879
Dep. SD	1.186	1.655	0.184	0.184

Notes: Binary post-treatment indicator: 1 in the event year and after. Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled treatments jointly; the four reported coefficients are β^M_{Inc}, β^M_{Dec}, β^{SD}_{Inc}, β^{SD}_{Dec}. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 15: Pooled Inc/Dec headline — municipality level (binary event)

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
Municipality rate (τ^M)					
$\Delta\tau > 0$ (Inc)	-0.117***	-0.099***	-0.000	-0.012***	0.014***
	(0.034)	(0.034)	(0.020)	(0.004)	(0.003)
$\Delta\tau < 0$ (Dec)	-0.141*	-0.049	0.038	-0.010**	0.000
	(0.074)	(0.038)	(0.058)	(0.005)	(0.002)
School district rate (τ^{SD})					
$\Delta\tau > 0$ (Inc)	0.031	0.062**	0.015***	0.001	-0.001
	(0.020)	(0.026)	(0.006)	(0.001)	(0.001)
$\Delta\tau < 0$ (Dec)	-0.007	0.023	-0.013	-0.002	-0.003
	(0.078)	(0.114)	(0.016)	(0.003)	(0.002)
N	218,677	239,023	242,722	239,033	242,722
R^2	0.964	0.959	0.981	0.823	0.945
Within R^2	0.001	0.002	0.001	0.001	0.001
Dep. mean	3.677	5.847	6.697	0.093	0.943
Dep. SD	1.934	1.887	1.299	0.076	0.066

Notes: Binary post-treatment indicator: 1 in the event year and after. Cluster-robust standard errors in parentheses. Each regression includes both rate-setters' direction-pooled treatments jointly; the four reported coefficients are β^M_{Inc}, β^M_{Dec}, β^{SD}_{Inc}, β^{SD}_{Dec}. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

Table 16: Pooled Inc/Dec headline — school district level (binary event)

Outcome	Log(work in own muni)	Log(in-commuters)	Log(employed residents)	Share live in muni	Share work outside
School district rate (τ^{SD})					
$\Delta\tau > 0$ (Inc)	0.089*	0.091**	0.129***	0.000	0.005
	(0.051)	(0.046)	(0.035)	(0.003)	(0.004)
$\Delta\tau < 0$ (Dec)	-0.007	-0.072***	-0.069***	0.008*	-0.006***
	(0.042)	(0.020)	(0.023)	(0.004)	(0.002)
N	51,440	51,474	51,474	51,474	51,474
R^2	0.986	0.987	0.987	0.953	0.977
Within R^2	0.004	0.007	0.006	0.006	0.005
Dep. mean	6.260	8.335	8.768	0.125	0.900
Dep. SD	1.298	1.202	0.909	0.068	0.076

Notes: Binary post-treatment indicator: 1 in the event year and after. Cluster-robust standard errors in parentheses. $^*p < 0.10$, $^{**}p < 0.05$, $^{***}p < 0.01$.

12 Sample Variant Robustness

References